INTERNAL HOM ON AN AFFINE SCHEME COMPUTES THE ALGEBRAIC HOM FOR A COHERENT FIRST ARGUMENT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0A88 (sources/stacks-duality/pinned-original-source.tex line 366).

Allowed standard prerequisites (editorial).

- Cohomology sheaves of the internal Hom on an affine scheme. Derived Categories of Schemes, Lemma labelled perfect-lemma-quasi-coherence-internal-hom, cited at proof lines 386-390. Identifies the presheaf whose sheafification is computed.
- Localisation of injective modules over a Noetherian ring. Dualizing Complexes, Lemma labelled dualizing-lemma-localization-injective-modules, cited at proof lines 396-399. Makes I otimes_A A_f injective.
- Hom from a finitely presented module commutes with localisation. Algebra, Lemma labelled algebra-lemma-hom-from-finitely-presented, cited at proof lines 406-410. Gives the last equality of the computation.

Source reference. Original lemma 0A6H: sources/stacks-chow/tags.txt, source lines 12845–12845. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 384 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A6I: sources/stacks-chow/tags.txt, source lines 12846–12846. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 393 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0583: sources/stacks-chow/tags.txt, source lines 6766–6766. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 404 of the counted proof. It

Date: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let A be a Noetherian ring and let $X = \mathrm{Spec}(A)$. Let K, L be objects of $D(A)$. If $K \in D_{Coh}(A)$ and L has finite injective dimension, then*

$$R\mathcal{H}om_{\mathcal{O}_X}(\widetilde{K}, \widetilde{L}) = R\widetilde{\mathrm{Hom}_A(K, L)}$$

in $D(\mathcal{O}_X)$.

Proof. We may assume that L is given by a finite complex $I^\bullet$ of injective A-modules. By induction on the length of $I^\bullet$ and compatibility of the constructions with distinguished triangles, we reduce to the case that $L = I[0]$ where I is an injective A-module. In this case, Derived Categories of Schemes, Lemma 0A6H, tells us that the nth cohomology sheaf of $R\mathcal{H}om_{\mathcal{O}_X}(\widetilde{K}, \widetilde{L})$ is the sheaf associated to the presheaf

$$D(f) \longmapsto \mathrm{Ext}^n_{A_f}(K \otimes_A A_f, I \otimes_A A_f)$$

Since A is Noetherian, the A_f-module $I \otimes_A A_f$ is injective (Dualizing Complexes, Lemma 0A6I). Hence we see that

$$\begin{aligned}
\mathrm{Ext}^n_{A_f}(K \otimes_A A_f, I \otimes_A A_f) &= \mathrm{Hom}_{A_f}(H^{-n}(K \otimes_A A_f), I \otimes_A A_f) \\
&= \mathrm{Hom}_{A_f}(H^{-n}(K) \otimes_A A_f, I \otimes_A A_f) \\
&= \mathrm{Hom}_A(H^{-n}(K), I) \otimes_A A_f
\end{aligned}$$

The last equality because $H^{-n}(K)$ is a finite A-module, see Algebra, Lemma 0583. This proves that the canonical map

$$R\widetilde{\mathrm{Hom}_A(K, L)} \longrightarrow R\mathcal{H}om_{\mathcal{O}_X}(\widetilde{K}, \widetilde{L})$$

is a quasi-isomorphism in this case and the proof is done. □